

Optimization

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Outline

Motivation

Entropy

Conditional Entropy and Mutual Information

Cross-Entropy and KL-Divergence



Let's work on this subject in our Optimization lecture

Cross Entropy

Cross Entropy: The expected number of bits when a wrong distribution Q is assumed while the data actually follows a distribution P

Objective function

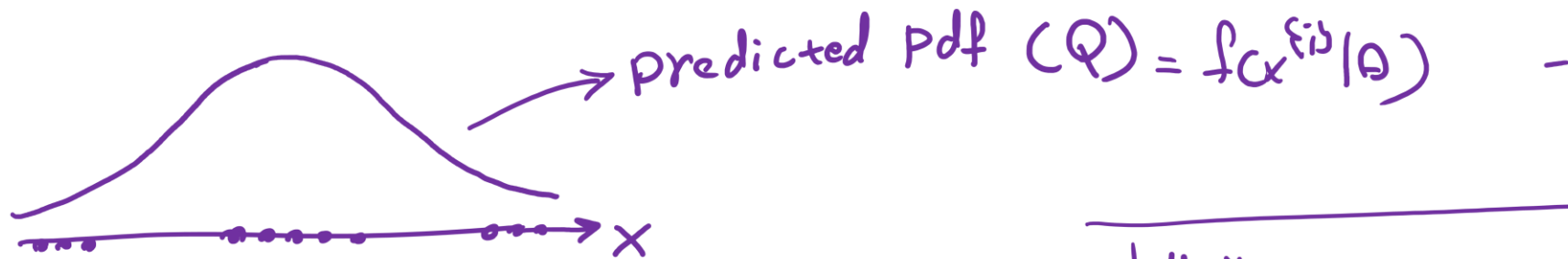
$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x) = \underbrace{H(P)}_{\text{actual pdf}} + \underbrace{KL[P][Q]}_{\text{predicted}}$$

This is because:

$$H(p, q) = \mathbb{E}_p[l_i] = \mathbb{E}_p \left[\log \frac{1}{q(x_i)} \right]$$

$$H(p, q) = \sum_{x_i} p(x_i) \log \frac{1}{q(x_i)}$$

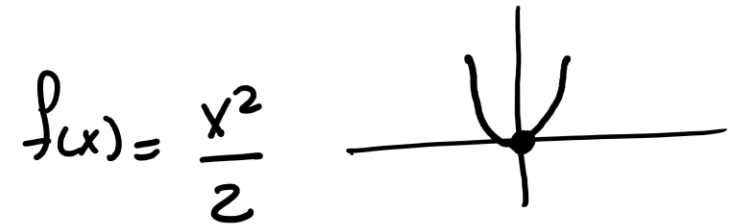
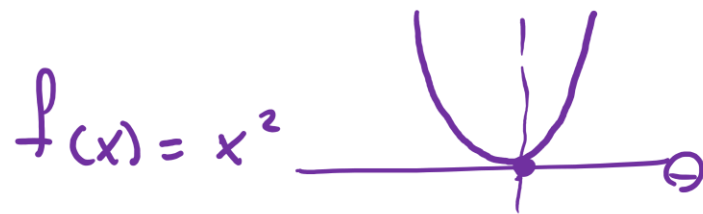
$$H(p, q) = - \sum_x p(x) \log q(x).$$



$$L(\theta | x) = \prod_{i=1}^N f(x^{i3} | \theta)$$

Objective function

$$\boxed{\text{Max } l(\theta | x) = \sum_{i=1}^N \log f(x^{i3} | \theta)}$$



Min negative log likelihood

$\frac{1}{N} \sum (\cdot)^{g(x)} = E[\cdot]^{g(x)}$

Min $\text{GLR } l(\theta | x) = - \sum_{i=1}^N \log f(x^{i3} | \theta)$

$$\text{Min } l(\theta | x) = \frac{1}{N} \sum \left(- \log f(x^{i3} | \theta) \right) = E \left[- \log f(x^{i3} | \theta) \right]$$

avg negative log likelihood = CE

$$= E \left[- \log Q \right]$$

$$E[-\log P + \log P - \log Q]$$

$$\log P - \log Q = \log \frac{P}{Q} \\ = -\log \frac{Q}{P}$$

$$E[A+B+C] = E[A] + E[B+C]$$

$$E[g(x)] = \sum \underline{p(x)} \underline{g(x)}$$

$$E[-\log P] + E[-\log \frac{Q}{P}]$$

↓

$$\sum p(x) * (-\log p(x)) + \sum p(x) * (-\log \frac{Q(x)}{p(x)})$$

$$\left(\sum p(x) \log_2 \frac{1}{p(x)} \right) - \sum p(x) \log \frac{Q(x)}{p(x)}$$

$$H(P) + KL[P][Q] = CE = \frac{1}{N} \sum -\log f(x^{(i)} | \theta)$$

$$-\cancel{\sum p(x) \log p(x)} - \sum p(x) \log Q(x) + \cancel{\sum p(x) \log p(x)}$$

$$-\sum p(x) \log Q(x) = CE$$

Labeling target values

Label encoding (ordinal) and One-hot encoding

$$X = \begin{bmatrix} \text{height} & \text{weight} & \text{age} & \dots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}_{n \times d} \rightarrow Y_a = \begin{bmatrix} \text{fish} \\ \text{cat} \\ \text{dog} \\ \vdots \end{bmatrix}_{n \times 1} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ \vdots \end{bmatrix} \rightsquigarrow \text{ML} \rightsquigarrow \hat{Y}_p = \begin{bmatrix} 0.5 \\ 1.2 \\ 1.8 \\ \vdots \end{bmatrix}$$

$$\text{fish} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \quad \text{cat} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \quad \text{dog} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$Y_a = \begin{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \\ \vdots \end{bmatrix} \rightsquigarrow \text{ML} \rightsquigarrow \hat{Y}_p = \begin{bmatrix} \begin{bmatrix} 78 & 20 & 10 \end{bmatrix} \\ \begin{bmatrix} 2 & 92 & 3 \end{bmatrix} \\ \begin{bmatrix} 2 & 3 & 80 \end{bmatrix} \\ \vdots \end{bmatrix} \rightsquigarrow \text{Softmax} \rightsquigarrow \begin{bmatrix} \begin{bmatrix} 0.8 & 0.1 & 0.1 \end{bmatrix} \\ \begin{bmatrix} 0.2 & 0.7 & 0.1 \end{bmatrix} \\ \begin{bmatrix} 0.3 & 0.1 & 0.6 \end{bmatrix} \\ \vdots \end{bmatrix}$$

$$\text{Max} \sum_{i=1}^N Y_a^{(i)} \cdot \hat{Y}_p^{(i)} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.1 \\ 0.1 \end{bmatrix} + \dots + \dots \Rightarrow \text{maximum possible is } = N$$

$$\text{Min} \sum_{i=1}^N \| Y_a^{(i)} - \hat{Y}_p^{(i)} \|_2^2 = \left[(1-0.8)^2 + (0-0.1)^2 + (0-0.1)^2 + \dots + \dots \right]$$

Why Cross entropy and not simply use dot product?

$\mathbb{R} \in$

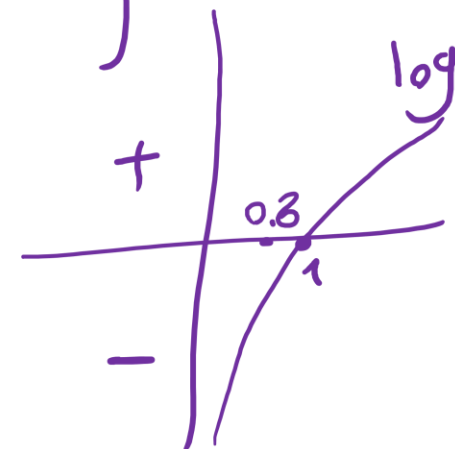
$$CE = - \sum_{i=1}^N y_a^{(i)} \log \hat{y}_p^{(i)}$$

$$y_a = \begin{bmatrix} 1 & 0 & 0 \\ \vdots & & \end{bmatrix}$$

$$\hat{y}_p = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ \vdots & & \end{bmatrix}$$

$$CE = - \left(\begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \log 0.8 \\ \log 0.1 \\ \log 0.1 \end{bmatrix} + \dots \right)$$

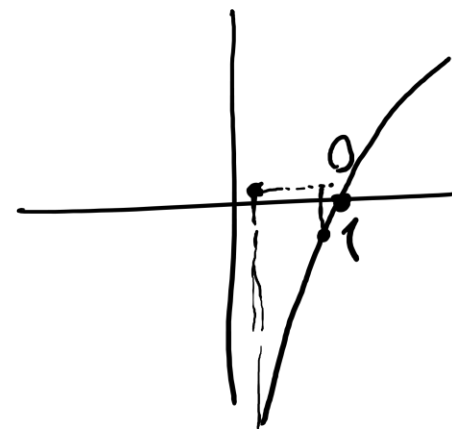
$y_a^{(i)}$ $\hat{y}_p^{(i)}$



① $-\sum y_a^{(i)} \hat{y}_p^{(i)}$

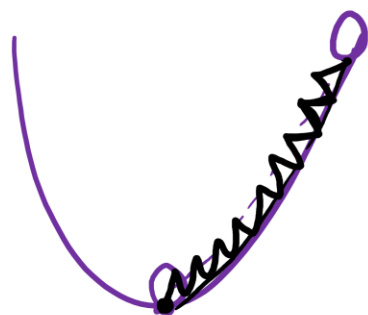


② $-\sum y_a^{(i)} \log \hat{y}_p^{(i)}$

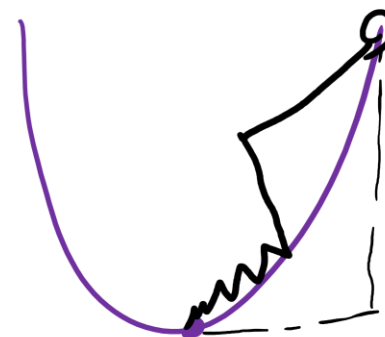


$$f(x) = x^2$$

$$f'(x) = 2x = 0$$



$$f(x) = x^2$$



Kullback-Leibler Divergence

Another useful information theoretic quantity measures the difference between two distributions.

$$\begin{aligned}\mathbf{KL}[P(S)||Q(S)] &= \sum_s P(s) \log \frac{P(s)}{Q(s)} \\ &= \underbrace{\sum_s P(s) \log \frac{1}{Q(s)}}_{\text{cross entropy}} - \mathbf{H}[P] = H(P, Q) - H(P)\end{aligned}$$

Excess cost in bits paid by encoding according to Q instead of P .

KL Divergence is
a **KIND OF**
distance
measurement

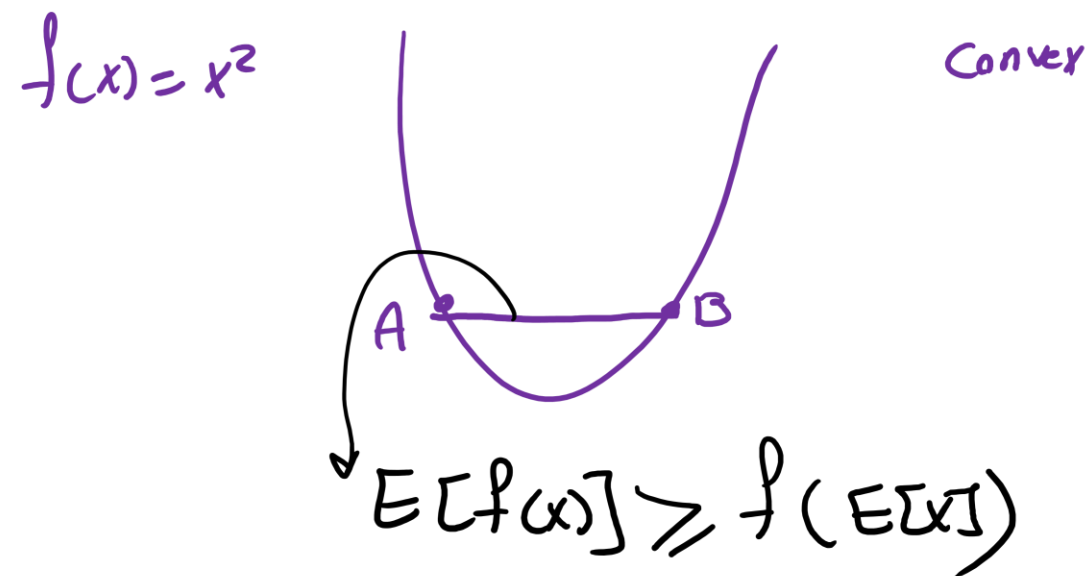
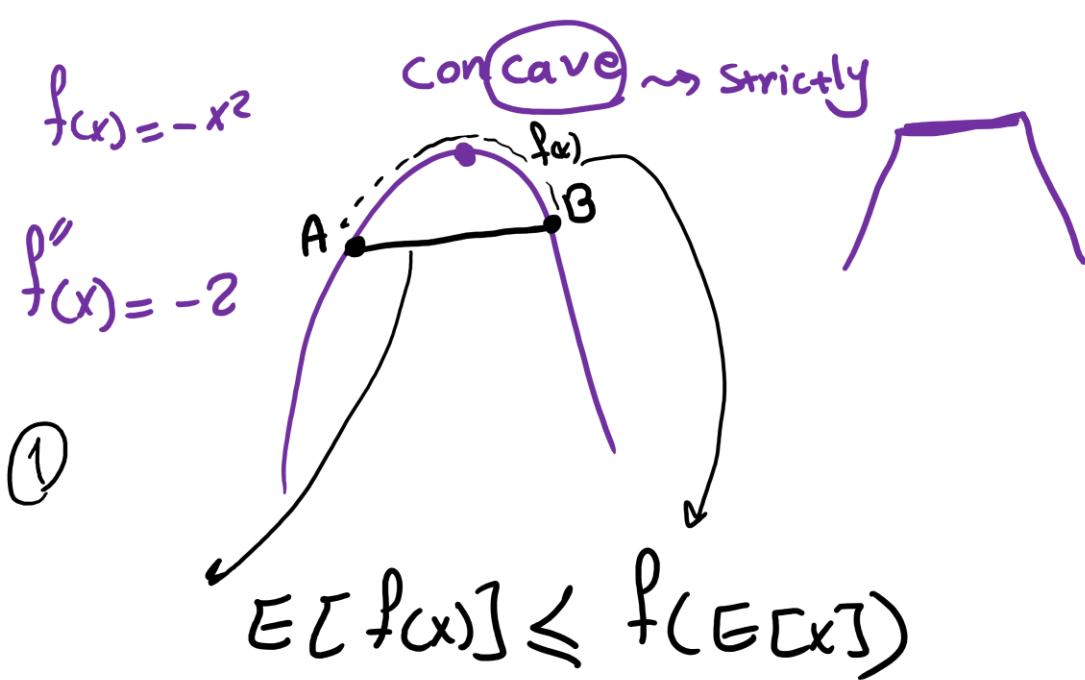
$$-\mathbf{KL}[P||Q] = \sum_s P(s) \log \frac{Q(s)}{P(s)}$$

log function is
concave or
convex?

$$\begin{aligned}\sum_s P(s) \log \frac{Q(s)}{P(s)} &\leq \log \sum_s P(s) \frac{Q(s)}{P(s)} && \text{By Jensen Inequality} \\ &= \log \sum_s Q(s) = \log 1 = 0\end{aligned}$$

So $\mathbf{KL}[P||Q] \geq 0$. Equality iff $P = Q$

When $P = Q$, $KL[P||Q] = 0$

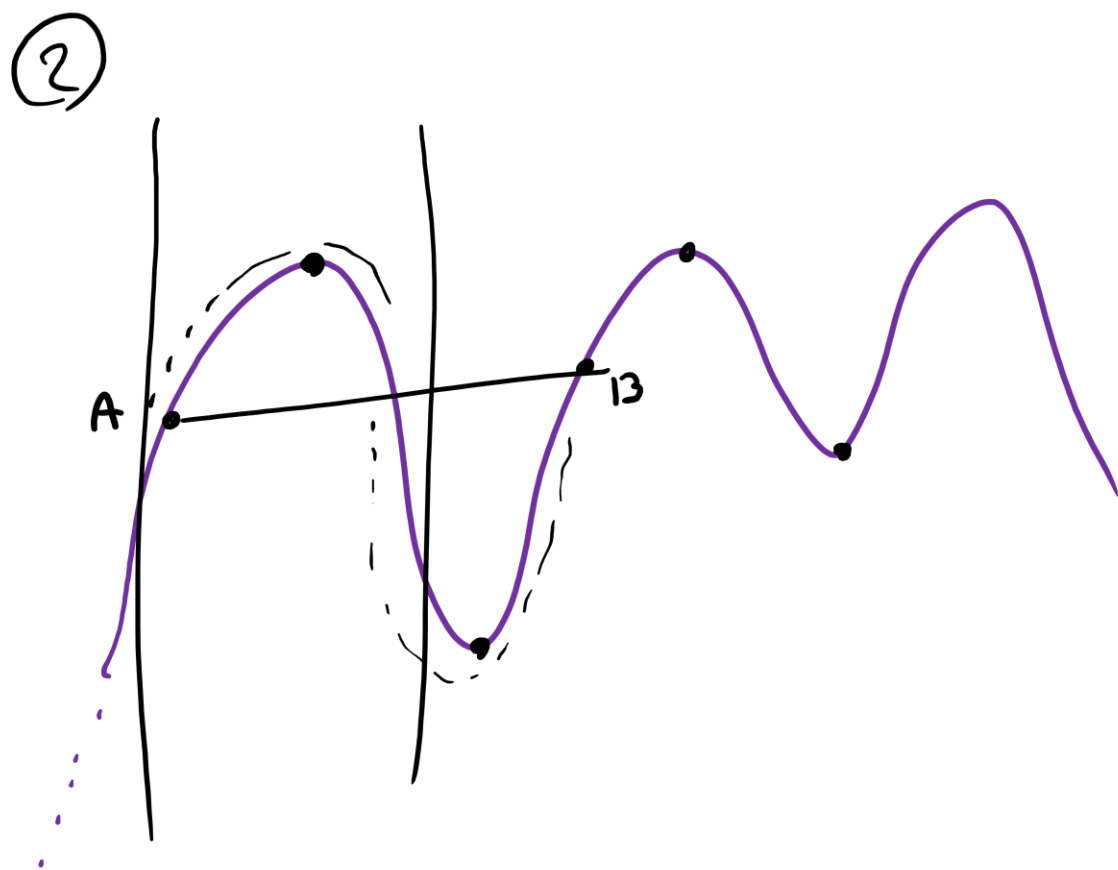


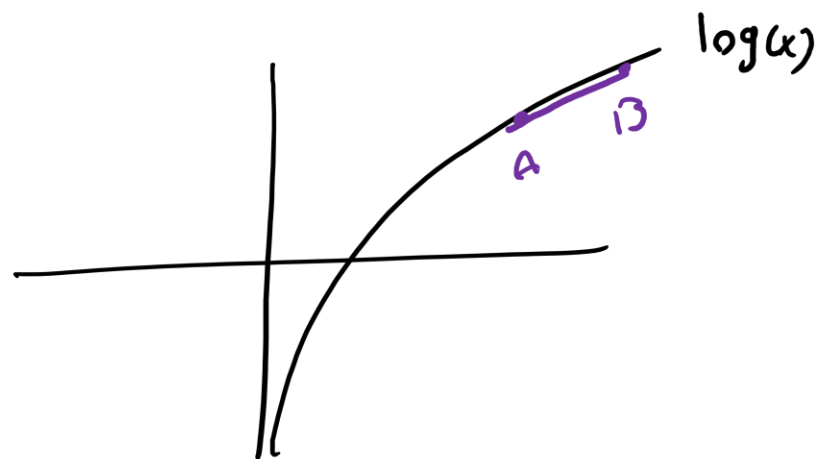
$f(x) = x^2$ $x = [1, 2]$
 $P(x) = [\frac{1}{2}, \frac{1}{2}]$

$E[f(x)] = \sum P(x) f(x) =$
 $P(x=1) f(x=1) + P(x=2) f(x=2)$
 $\frac{1}{2} * 1 + \frac{1}{2} * 4 = 2.5$

$f(E[x]) = f(E[x]=1.5) = (1.5)^2$

$E[x] = \sum P(x) * x = \frac{1}{2} * 1 + \frac{1}{2} * 2 = 1.5$





$$E[f(x)] \leq f(E[x]) \Rightarrow E[\log x] \leq \log(E[x])$$

$$f(x) = \log x$$

$$-KL[P][Q] = \sum p(x) \log \left(\frac{q(x)}{p(x)} \right) = \sum p(x) \log q(x)$$

$$E[\log q(x)] \leq \log E[q(x)]$$

$$\begin{aligned} &\leq \log \sum p(x) q(x) \\ &\leq \log \sum \cancel{p(x)} \frac{q(x)}{\cancel{p(x)}} \\ &\leq \log \left(\sum q(x) \right) \end{aligned}$$

$$-KL[P][Q] \leq \log 1 = 0 \Rightarrow KL[P][Q] \geq 0$$

Objective function $\leftarrow f(x, y)$

S.t. $g(x, y) = 0 \rightsquigarrow$ Equality constraint

S.t. $h(x, y) \leq 0 \rightsquigarrow$ Inequality constraint (KKT conditions)

$$f''(x, y) \rightsquigarrow H = \begin{bmatrix} \frac{\partial^2 f(x, y)}{\partial x^2} & \frac{\partial^2 f(x, y)}{\partial x \partial y} \\ \frac{\partial^2 f(x, y)}{\partial y \partial x} & \frac{\partial^2 f(x, y)}{\partial y^2} \end{bmatrix}$$

① Linear programming

$$f(x, y) = x + y$$

$$\text{s.t. } x + y = 0$$

② Quadratic //

$$f(x, y) = x^2 + y^2$$

$$\text{s.t. } x + y = 0$$

③ non-linear //

$$f(x, y) = x^3 + y^2$$

$$\text{s.t. } x^2 + y = 0$$

$$f(m,s) = 6m^2 + 3s^2$$

M = # hours you study ML per day

S = " " you sleep " "

$$\frac{\partial f(m,s)}{\partial m} = 0 \Rightarrow 12m = 0 \Rightarrow m = 0$$

$$\frac{\partial f(m,s)}{\partial s} = 0 \Rightarrow 0 + 6s = 0 \Rightarrow s = 0$$

→ Objective function

$\lambda, \alpha, \beta \rightsquigarrow$ Lagrange multiplier

$$f(M, S) = 6M^2 + 3S^2$$

s.t.

$$M + S = 24$$

Constraint function

$$g(M, S) = M + S - 24$$

$$L(M, S, \lambda) = f(M, S) - \lambda g(M, S) = 6M^2 + 3S^2 - \lambda(M + S - 24)$$

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow 0 + 0 - 1(M + S - 24) = 0 \Rightarrow M + S = 24 \Rightarrow \frac{\lambda}{12} + \frac{\lambda}{6} = 24$$

$$\frac{\partial L}{\partial M} = 0 \Rightarrow 12M + 0 - \lambda = 0 \Rightarrow M = \frac{\lambda}{12} = \frac{96}{12} = 8 \Rightarrow \lambda = 96$$

$$\frac{\partial L}{\partial S} = 0 \Rightarrow 6S - \lambda = 0 \Rightarrow S = \frac{\lambda}{6} = \frac{96}{6} = 16$$

$$8 + 16 = 24$$

$$f(u,s)$$

$$\text{s.t. } g(u,s) = 0 \rightarrow \lambda_1$$

$$\text{s.t. } h(u,s) = 0 \rightarrow \lambda_2$$

$$L(u,s,\lambda_1,\lambda_2) = f(u,s) - \lambda_1 g(u,s) - \lambda_2 h(u,s)$$

$$h = f(m,s) - \lambda g(m,s)$$

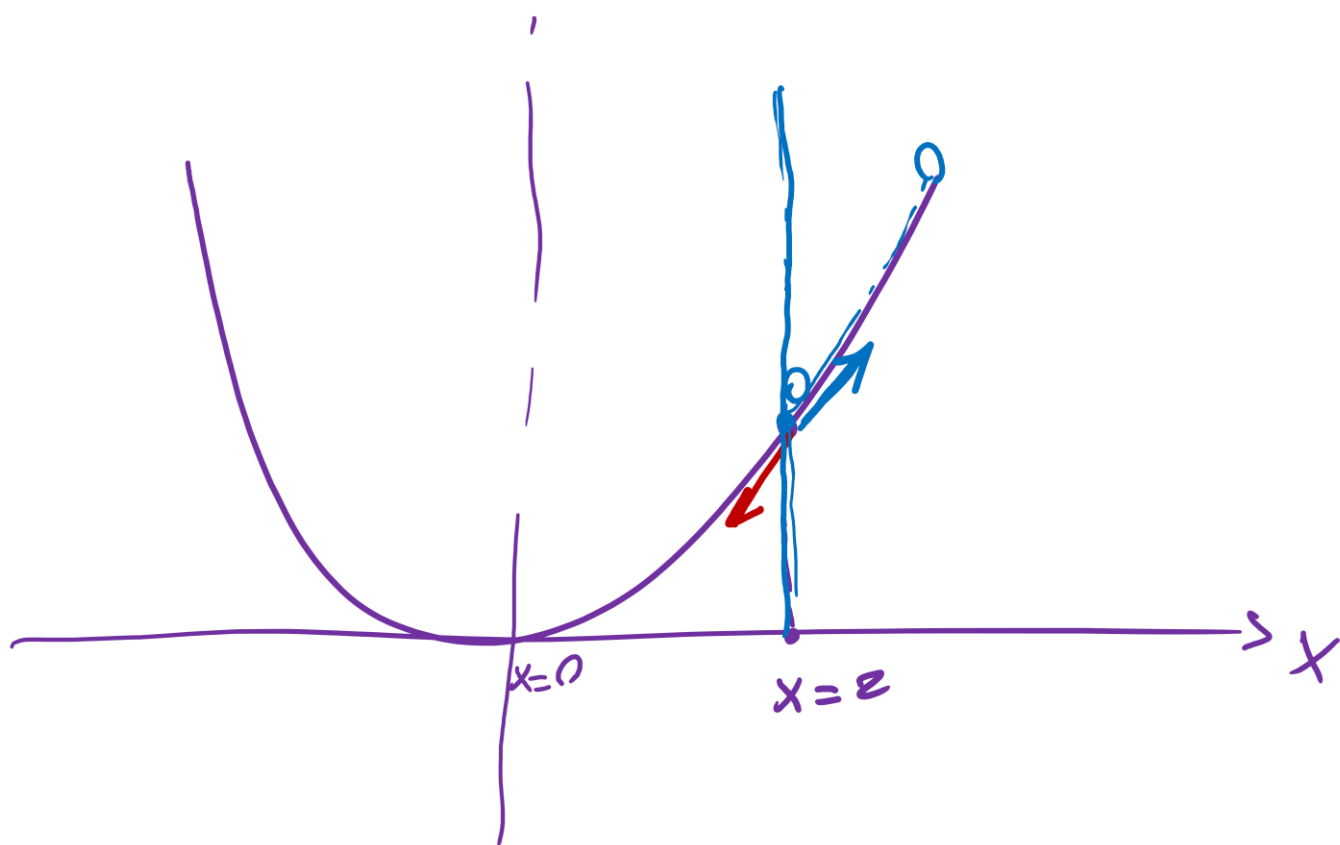
$$\nabla h = 0 \Rightarrow \nabla (f(m,s) - \lambda g(m,s)) = 0$$

$$\Rightarrow \nabla f(m,s) - \lambda \nabla g(m,s) = 0 \Rightarrow \nabla f(m,s) = \cancel{\lambda} \nabla g(m,s)$$

$$\nabla f(m,s) \approx \nabla g(m,s)$$

↓

$$\nabla f(m,s) = \lambda \nabla g(m,s)$$



$$f(M, S) = 6M^2 + 3S^2$$

$$M + S \leq 24 \Rightarrow g(M, S) = M + S - 24 \leq 0$$

Active $g(M, S) = 0$

Inactive $g(M, S) < 0$

$$L(M, S, \lambda) = (6M^2 + 3S^2) + \lambda(M + S - 24)$$

KKT conditions

① Stationary condition $\nabla L(M, S, \lambda) = 0$

② primal feasibility: $g(M, S) \leq 0$

③ Dual feasibility: $\lambda \geq 0$

④ complementary slackness: $\lambda g(M, S) = 0$

$$\rightarrow g(M, S) = 0 \leadsto \lambda \geq 0$$

$$\rightarrow g(M, S) \neq 0 \leadsto \lambda = 0$$

$$f(m,s) = \frac{1}{2} m^2 + \frac{1}{2} s^2$$

Convex $\textcircled{\text{min}}$ 

Primal form

$$\text{s.t. } m+s = 24$$

$$\frac{\partial L}{\partial m} = 0 \Rightarrow m = s$$

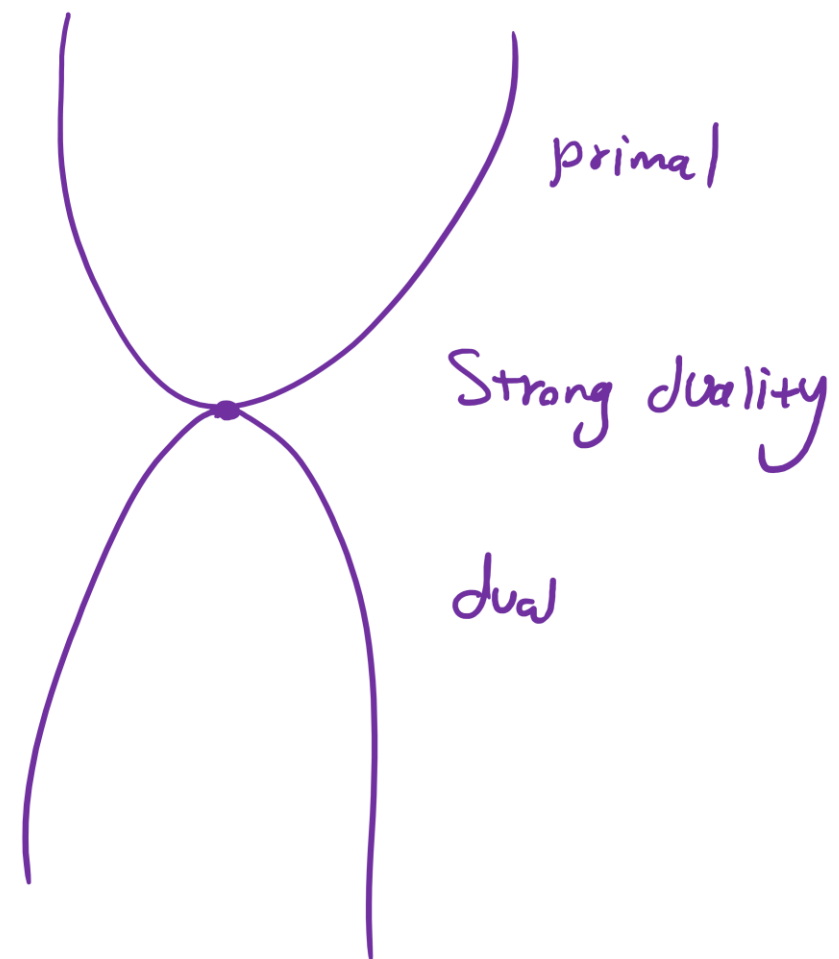
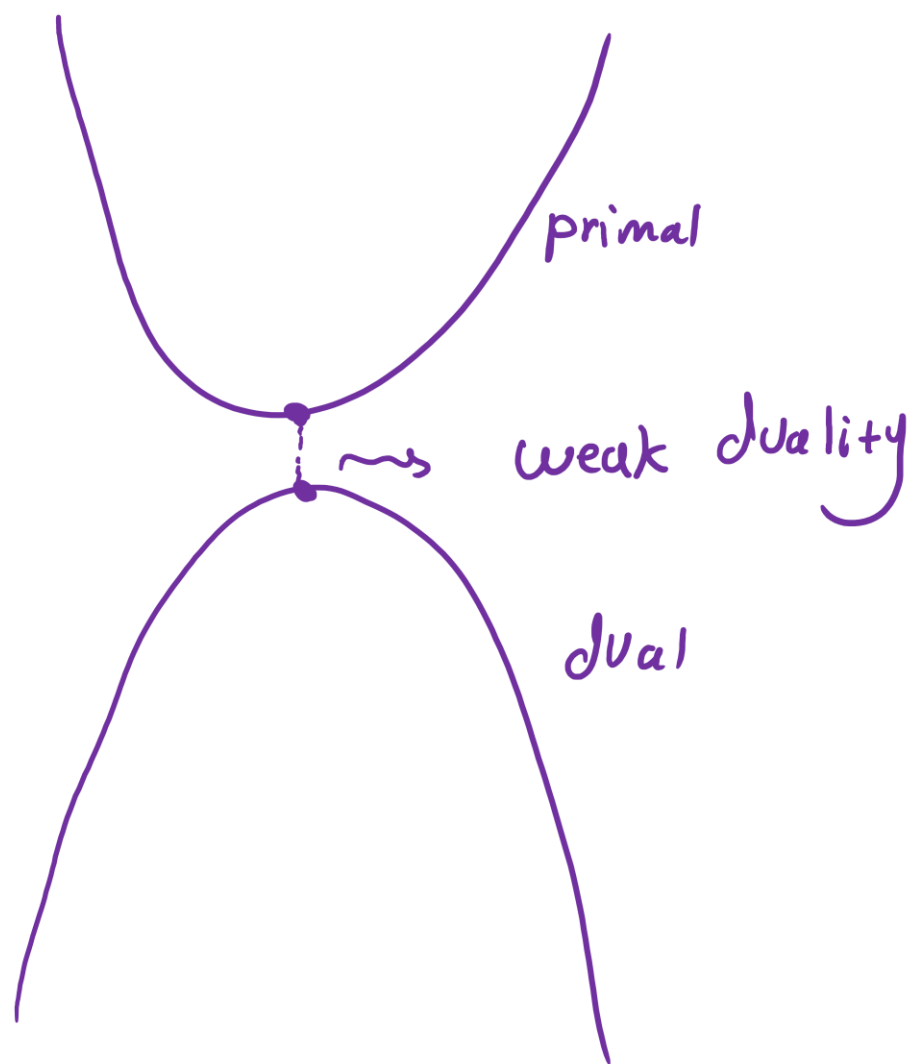
$$h(m,s,\lambda) = \frac{1}{2} m^2 + \frac{1}{2} s^2 - \lambda (m+s-24) \Rightarrow \frac{\partial h}{\partial s} = 0 \Rightarrow s = \lambda$$

$$L(\lambda) = \frac{1}{2} \lambda^2 + \frac{1}{2} \lambda^2 - \lambda (2\lambda - 24) = \lambda^2 - 2\lambda^2 + 24\lambda = \underbrace{-\lambda^2 + 24\lambda}_{\text{Max}}$$

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow \lambda = 12$$

Max



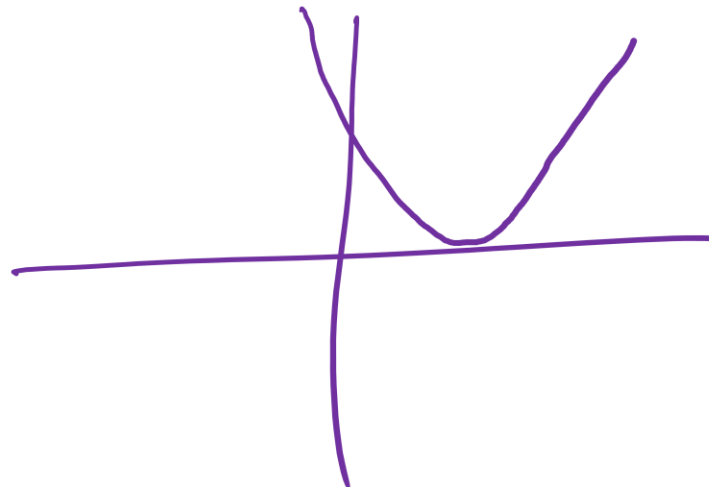


$$f(x) = e^x + x^2$$

$$f'(x) = (e^x + 2x) = 0$$

$$f''(x) = e^x + 2 > 0 \text{ convex}$$

$$e^x + 2x = 0$$

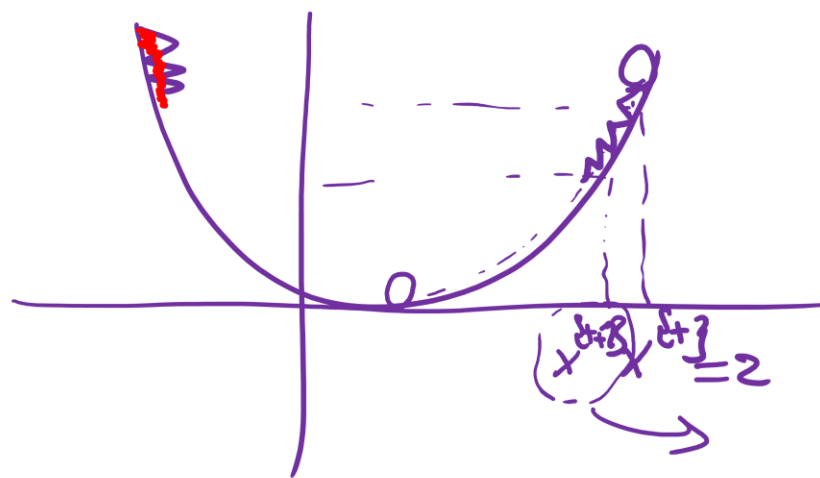


GD : gradient descent

GA : gradient ascent

$$x^{t+1} \leftarrow x^t - \alpha \frac{\partial f(x)}{\partial x}$$

$$x^{t+1} \leftarrow x^t - \alpha (e^{x^t} + 2x^t) = 2 - 0.02(e^2 + 4)$$



① Quotient rule: $\left(\frac{f}{g}\right)' = \frac{f'g - g'f}{g^2}$

② Product rule: $(fg)' = f'g + g'f$

③ Chain rule: $(v^n)' = nv'v^{n-1}$

1	0	-1
1	0	-1
1	0	-1

~~1~~

0	1	2
2	1	3
1	2	1

二

0	0	-2
2	0	-3
1	0	-1

$$\text{Sum} \left(\begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array} \right) =$$

$$0 + 0 + -2 + 2 + \dots =$$

$$-3$$

0	1	2	4	5	6	0	2
2	1	3	8	9	255	72	83
1	2	1	4	79	65	53	33
43	97	15	67	104	77	163	43
36	173	13	76	205	89	179	34
63	163	113	86	209	91	185	98
84	153	123	96	134	101	196	121
96	143	133	79	135	103	216	211

[illegible]

